ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

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ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 8

Full Practice Test • Test #8

Geometry, 3D solids, tangents, logic & coordinate geometry. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #8

Practice Questions

50 QUESTIONS

1 Each of these 5 clues is a true statement:
Clue 1: George, Wendy, and Juanita did not bring their lunches.
Clue 2: George ate the last piece of candy if and only if Juanita brought her lunch.
Clue 3: If Wendy did not bring her lunch, then Juanita did not eat the last piece of candy.
Clue 4: If Dashon ate the last piece of candy, then Wendy brought her lunch.
Clue 5: Only one of these people ate the last piece of candy.

Use these clues to determine who ate the last piece of candy.

Clue #	Statement
Clue 1	George, Wendy, and Juanita did not bring their lunches.
Clue 2	George ate the last piece of candy if and only if Juanita brought her lunch.
Clue 3	If Wendy did not bring her lunch, then Juanita did not eat the last piece of candy.
Clue 4	If Dashon ate the last piece of candy, then Wendy brought her lunch.
Clue 5	Only one of these people ate the last piece of candy.

(A) Dashon
(B) George
(C) Juanita
(D) Wendy

2 Which of the following is $3(4-x^2)+10$ in simplified form?

(A) $22-x^2$
(B) $22-3x^2$
(C) $12-x^2$
(D) $12-3x^2$

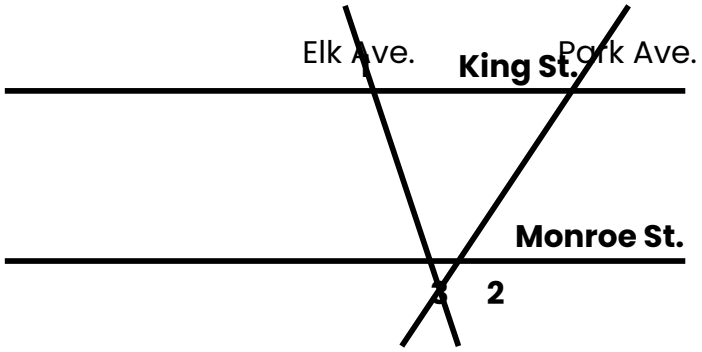
3 Mr. Brady's 9th grade science class grows a type of bacterium in a petri dish. The table shows the number of bacteria in the dish at different times. If the pattern continues, how many bacteria will be in the dish after 10 hours?

Time in hours	Number of bacteria
0	1
1	2
2	4
3	8

(A) 256
(B) 512
(C) $1,024$
(D) $2,048$

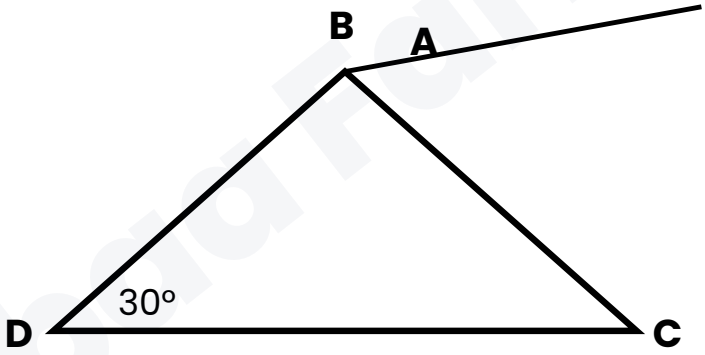
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4 This map shows the intersections of four roads. King Street and Monroe Street are parallel. Elk Avenue intersects King Street such that $m\angle 1 = 71°$. Park Avenue intersects Monroe Street such that $m\angle 2 = 43°$. What is $m\angle 3$?



A $38°$
B $56°$
C $66°$
D $94°$

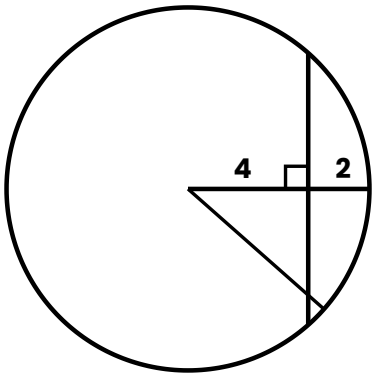
5 In ΔBCD, $m\angle BDC = 30°$, if the measure of exterior $\angle ABD$ is twice $m\angle DCB$, what is $m\angle ABD$?



A $30°$
B $60°$
C $105°$
D $120°$

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6 The chord of this circle is perpendicular to a radius of the circle. The chord divides the radius into segment 2 cm and 4 cm long. What is the length, in centimeters, of the chord?



(A) $2\sqrt{5}$
(B) $2\sqrt{13}$
(C) $4\sqrt{5}$
(D) 10

7 What is the square root of $25 \cdot 3^{16}$?

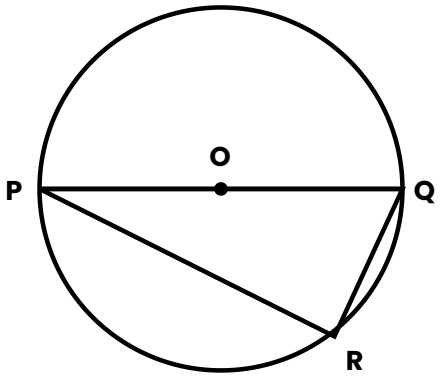
(A) $5 \cdot 3^4$
(B) $5 \cdot 3^8$
(C) $5 \cdot 3^{14}$
(D) $5 \cdot 3^{16}$

8 A line has a slope of -2 and passes through the point $(4, -5)$. What is the equation of this line in standard form?

(A) $2x + y = 3$
(B) $2x + y = 13$
(C) $2x + y = -6$
(D) $2x + y = -13$

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9 In the figure, $\overline{PQ}$ is a diameter of circle O, and R lies on the circle. If QR is 1 unit and the radius of the circle is $\frac{5}{2}$ units, how many units is PR?



(A) $\frac{\sqrt{21}}{2}$
(B) 4
(C) $\sqrt{24}$
(D) $\sqrt{26}$

10 A student gave this proof that $3^0 = 1$:
What is the justification for the fourth step of the proof?

Step	Justification
$3^0 = 3^{(2-2)}$	Subtraction fact
$3^{(2-2)} = 3^2 / 3^2$	Quotient property of exponents
$3^2 / 3^2 = 3^2 \cdot (1 / 3^2)$	Definition of division
$3^2 \cdot (1 / 3^2) = 1$	_____
$3^0 = 1$	Transitive property of equality

(A) Identity property for multiplication
(B) Inverse property of multiplication
(C) Identity property for addition
(D) Multiplication property of equality

11 What is the standard form equation of a circle with a diameter that has endpoints of $(-1, 6)$ and $(5, 2)$?

(A) $(x-2)^2 + (y-4)^2 = 13$
(B) $(x+2)^2 + (y+4)^2 = 13$
(C) $(x-2)^2 + (y+4)^2 = 26$
(D) $(x+2)^2 + (y-4)^2 = 26$

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12 Jordan is considering purchasing two posters. Each dimension of one poster is three times the corresponding dimension of the other poster. How many times greater is the area of the larger poster than the smaller one?

A) 3
B) 6
C) 9
D) 12

13 If Event 1 occurs before Event 2, which of the following pairs of events are independent?

A) Event 1: Toss a coin. Event 2: Toss the same coin again.
B) Event 1: Select a card from a deck of 52 cards. Event 2: Select another card without replacing.
C) Event 1: It begins raining outside at 1:15 p.m. Event 2: The sidewalk outside is wet at 1:20 p.m.
D) Event 1: Select a marble from a bag. Event 2: Select a second marble without replacing.

14 Solve $2|3x - 8| - 3 = 11$

A) $x = 4$ or $x = \frac{4}{3}$
B) $x = \frac{20}{3}$ or $x = -\frac{4}{3}$
C) $x = 8$ or $x = -\frac{8}{3}$
D) $x = 5$ or $x = \frac{1}{3}$

15 Simplify $\frac{\sqrt{8x^3}}{\sqrt{27x}}$ for $x > 0$

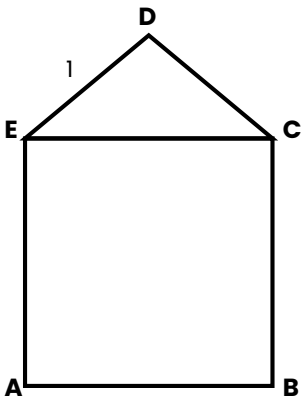
A) $\frac{4x}{3}$
B) $\frac{2x}{3}$
C) $\frac{2x\sqrt{6}}{9}$
D) $\frac{2x\sqrt{6}}{3}$

16 Completely factor $4x^2 - 12xy + 9y^2$

A) $(2x - 3y)^2$
B) $(2x + 9y)^2$
C) $(4x - 9y)^2$
D) $4(x - 3y)^2$

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17 This is a scale model of a house. Triangle EDC is an isosceles right triangle, and $ABCE$ is a square. If $DE = 1$, what is the perimeter of $ABCDE$?



(A) 5
(B) 6
(C) $2 + 3\sqrt{2}$
(D) $4\sqrt{2}$

18 For the equation $x^2 - 2a = 0$, what value(s) of a will result in an equation with two distinct real roots?

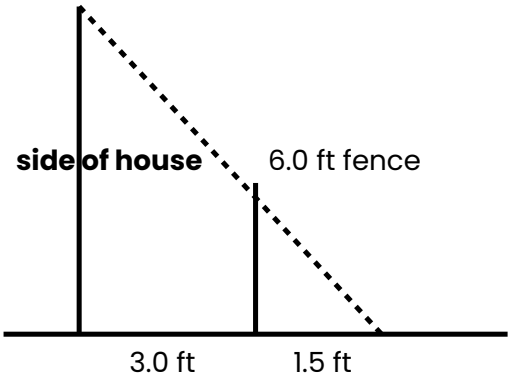
(A) $a < -2$
(B) $a > 0$
(C) $a = 0$
(D) $a < 0$

19 The area of a triangle is found using $A = \frac{1}{2}bh$, where b is the base and h is the height. Which formula yields the height of a triangle?

(A) $h = \frac{2A}{b}$
(B) $h = \frac{A}{2b}$
(C) $h = \frac{AB}{2}$
(D) $h = \frac{b}{2A}$

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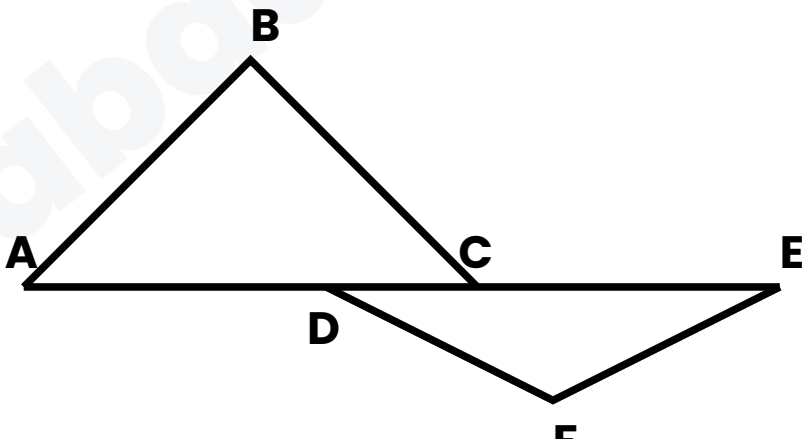
20 A ladder is leaning against the side of house. The base of the ladder is 1.50 ft from the base of a 6.00 ft tall fence. The fence is 3.00 ft from, and parallel to, the side of the house. If the ladder touches the top of the fence, how many feet does the ladder reach up the house's side?



A) 18 ft
B) 19 ft
C) 20 ft
D) 21 ft

21 Given $\Delta BAC \cong \Delta FED$, C and D are on $\overline{AE}$, prove $\overline{AB} \parallel \overline{EF}$. Which is the reason for statement 3?

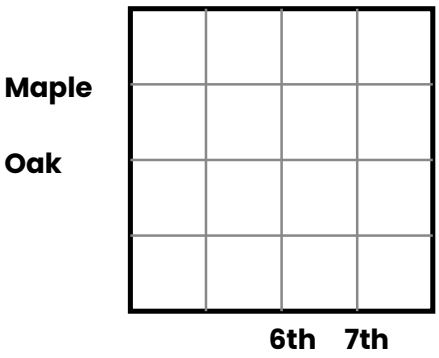
Statements	Reasons
1. ΔBCA ≅ ΔFED	Given
2. ∠BAE ≅ ∠FEA	CPCTC
3. AB || EF	_____



A) If two lines cut by a transversal form a pair of congruent alternate interior angles, then the two lines are parallel.
B) If two lines cut by a transversal form a pair of congruent alternate exterior angles, then the two lines are parallel.
C) Corresponding parts of congruent triangles are congruent.
D) Congruent corresponding sides of congruent triangles are parallel.

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22 The town of Newton purchased a new emergency siren that can be heard within $\frac{1}{2}$ mile radius. Assuming the square blocks are all the same size, at which corner should the police install the siren so that it is within $\frac{1}{2}$ mile of any street corner in Newton?



A Maple and 6th
B Maple and 7th
C Oak and 6th
D Oak and 7th

23 Jade bought 2 similar cylindrical planters. Both the diameter and height of the larger one were 3 times that of the smaller one. It took 1 bag of soil to fill the small planter. How many bags of soil does Jade need to fill the larger planter?

A 3
B 9
C 10
D 27

24 Find the difference: $(x^2 + 4x + 4) - (5x^2 - x - 2)$

A $4x^2 + 3x + 2$
B $4x^2 + 3x + 6$
C $-4x^2 + 5x + 6$
D $-4x^2 + 5x + 2$

25 Which of the following expressions is a simplified form of this expression? $(\sqrt{16x^6})^2(\sqrt{4x^4})^2$

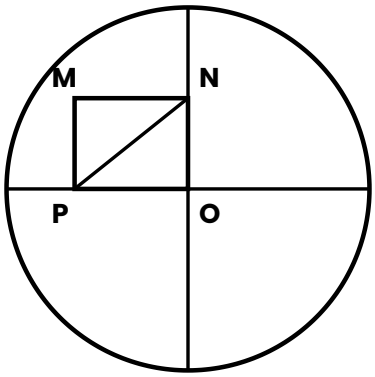
A $8x^6$
B $64x^{10}$
C $64x^{24}$
D $4,096x^{20}$

26 The equation $d = 16t^2$ models the distance (d), in feet, that a free-falling object travels in t seconds. According to this model, how many seconds does it take a free-falling object to travel 162 ft?

A $\frac{9}{4}$
B $\frac{9\sqrt{2}}{2}$
C $\frac{9\sqrt{2}}{4}$
D $\frac{9\sqrt{2}}{16}$

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27 Circle O has a radius of 2 cm, and $MNOP$ is a rectangle. Which reason justifies $PN = 2$ cm?

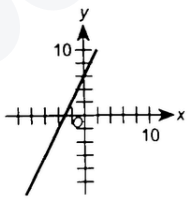


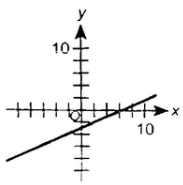
A The diagonals of a rectangle are congruent.
B Both pairs of opposite sides of a rectangle are parallel.
C The sum of the angles in a rectangle is $360°$.
D Pythagorean theorem

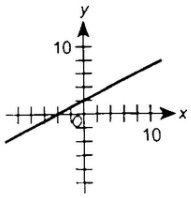
28 If $f(x) = x^2 - 3x - 1$ find $f(-2)$.

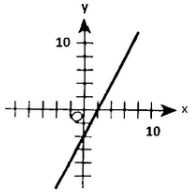
A -11
B -3
C 1
D 9

29 A function $y = f(x)$, consists of the set of all ordered pairs such that the first member of the ordered pair is 3 less than $\frac{1}{2}$ the second member. Which of the following is the graph of this function?

A


C


B


D


A Line with y-intercept 6 and slope 2
B Line with y-intercept 3 and slope $\frac{1}{2}$
C Line with y-intercept -3 and slope $\frac{1}{2}$
D Line with y-intercept -6 and slope 2

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30 Which set of measurements could be the side lengths of an obtuse triangle?

A 3 in, 4 in, 4 in

B 5 in, 12 in, 13 in

C 6 in, 7 in, 12 in

D 6 in, 8 in, 9 in

31 What is the equation in slope-intercept form of the line that passes through the point $(2, -2)$ and is perpendicular to the line represented by $y = \frac{2}{5}x + 2$?

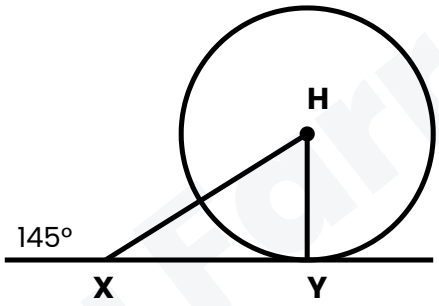
A $y = \frac{5}{2}x - 7$

B $y = \frac{5}{2}x + 7$

C $y = -\frac{5}{2}x + 3$

D $y = -\frac{5}{2}x - 3$

32 In the circle, H is the center, and $\overline{XY}$ is tangent to the circle, what is the measure of $\angle XHY$?



A $25°$

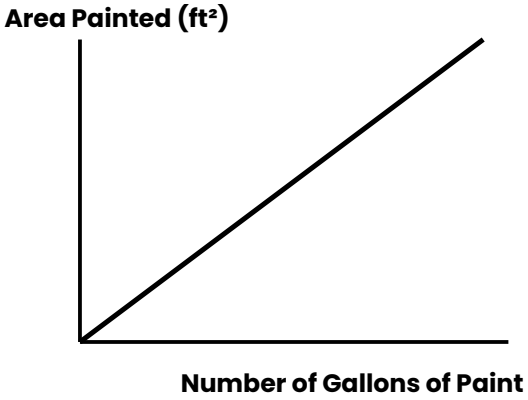
B $55°$

C $90°$

D $125°$

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33 The graph shows the linear relationship between the maximum area in square feet that can be painted and the number of gallons of paint used. Which of these represents the rate of change of the maximum area painted with respect to the number of gallons of paint used?



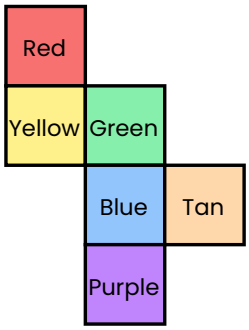
A $200\ \text{ft}^2/\text{gal}$
B $\frac{1}{200}\ \text{ft}^2/\text{gal}$
C $400\ \text{ft}^2/\text{gal}$
D $\frac{1}{400}\ \text{ft}^2/\text{gal}$

34 Two customers went to a post office to buy postcards and large Envelopes. Each postcard costs the same amount, and a large envelope costs the same amount.
• The first customer paid \$12 for 14 postcards and 5 large envelopes.
• The second customer paid \$24.80 for 10 postcards and 15 large envelopes.
What was the cost in dollars of each large envelope?

A \$1.42
B \$0.35
C \$1.15
D \$0.63

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35 Imagine that this net will be folded into a cube. Which color will not be adjacent to the red face?



(A) Purple
(B) Green
(C) Blue
(D) Tan

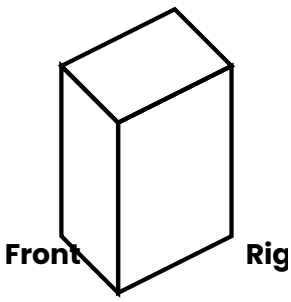
36 Which function is equivalent to $f(x) = -4(x + 7)^2 - 6$?

(A) $f(x) = -4x^2 - 56x - 202$
(B) $f(x) = -4x^2 + 14x + 43$
(C) $f(x) = -4x^2 - 56x - 172$
(D) $f(x) = -4x^2 + 190$

37 Which expression is a factor of $21x^2 + 13x - 20$?

(A) $3x - 4$
(B) $7x - 5$
(C) $7x + 4$
(D) $3x + 5$

38 Which set shows the front, right, and top views of this three-dimensional figure?



(A) View Set A
(B) View Set B
(C) View Set C
(D) View Set D

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39 In a sequence of numbers $a_3 = 0$, $a_4 = 6$, $a_5 = 12$, $a_6 = 18$, and $a_7 = 24$. Based on this information, which equation can be used to find n^{th} term in the sequence, a_n?

A $a_n = -6n + 18$
B $a_n = -18n + 6$
C $a_n = 6n - 18$
D $a_n = 18n - 6$

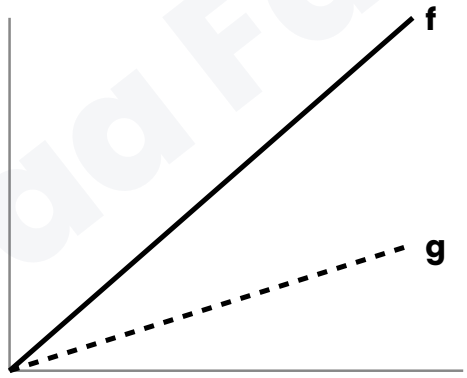
40 Kayle was cutting right triangles from wood to one of an art project. Two of the right triangles she cut are shown below. If $\Delta ABC \sim \Delta DEF$ with right angle B and E, $BC = 15$ cm and $AC = 17$ cm. What is the measure of $\angle F$ to the nearest degree?

A $28°$
B $41°$
C $62°$
D $88°$

41 Which equation represents the line passing through the points $(3, 2)$ and $(-9, 6)$?

A $x - 3y = 9$
B $x + 3y = 9$
C $3x - y = -9$
D $3x + y = 9$

42 The graph of linear function f and g are shown on the grid. Which function is best represented by the graph of g?



A $g(x) = f(x) - 4$
B $g(x) = \frac{1}{3}f(x)$
C $g(x) = f(x) - 2$
D $g(x) = 3f(x)$

43 In the diagram below of ΔABC, $\angle ABC$ is a right angle, $AC = 12$, $AD = 8$ and altitude $\overline{BD}$ is drawn. What is the length of $\overline{BC}$?

A $4\sqrt{2}$
B $4\sqrt{3}$
C $4\sqrt{5}$
D $4\sqrt{6}$

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44 A circle whose center is the origin passes through the point $(-5, 12)$. Which point also lies on the circle?

A $(10, 3)$
B $(-12, 13)$
C $(11, 2\sqrt{12})$
D $(-8, 5\sqrt{21})$

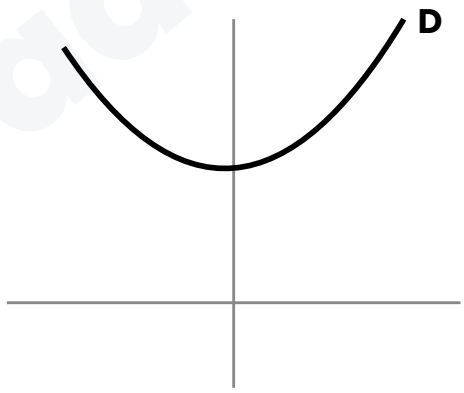
45 What is the contrapositive of the statement below?
"If today is Friday, then tomorrow is Saturday"

A If tomorrow is not Saturday, then today is not Friday.
B If today is Saturday, then tomorrow is not Friday.
C If tomorrow is Saturday, then today is Friday.
D If today is not Friday, then tomorrow is not Saturday.

46 Which statement about the graph of $y = 8(0.25)^x$ is true?

A The coordinates of the x-intercept are $(0.25, 0)$
B The coordinates of the y-intercept are $(0, 8)$
C The equation of the asymptote is $x = 0$
D The graph includes the point $(2, 1)$

47 Which graph represents a function?



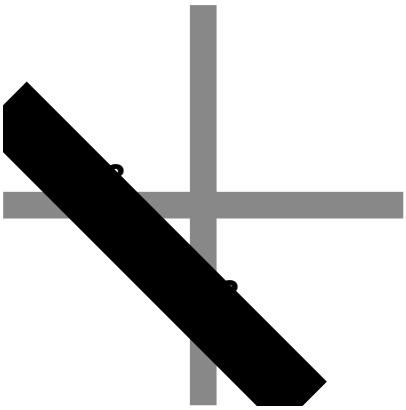
A Graph A
B Graph B
C Graph C
D Graph D

48 Two semicircles are congruent and are collinear. How many points and lines of symmetry do they have, if any?

A 1 Point, 2 line
B 2 point, 1 line
C 1 point, 1 line
D 0 point, 0 line

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49 In $f(x) = y$, a point with coordinates x, y lies. If the first member of the coordinates lies 3 less than the second member, which graph represents $f(x)$?



(A) Line with y-intercept -3
(B) Line with x-intercept -3
(C) Line with x-intercept 3
(D) Line with y-intercept 3 and x-intercept -3

50 Triangle JGR is similar to triangle MST. Which statement is not always true?

(A) $\angle J \cong \angle M$
(B) $\angle G \cong \angle T$
(C) $\angle R \cong \angle T$
(D) $\angle G \cong \angle S$

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TEST #8 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	D	26	C
2	B	27	A
3	C	28	D
4	C	29	A
5	B	30	C
6	C	31	C
7	B	32	B
8	A	33	C
9	C	34	A
10	B	35	C
11	A	36	A
12	C	37	B
13	A	38	D
14	D	39	C
15	C	40	A
16	A	41	B
17	C	42	B
18	B	43	B
19	A	44	C
20	A	45	A
21	A	46	B
22	A	47	D
23	D	48	A
24	C	49	D
25	B	50	B

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TEST #8 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: D** Wendy

Step 1: Clue 1: George, Wendy and Juanita did not bring lunch.
Step 2: Clue 2: George ate the candy only if Juanita brought lunch — she didn't, so George did not eat it.
Step 3: Clue 3: Wendy didn't bring lunch, so Juanita did not eat it. Clue 4: if Dashon ate it, Wendy brought lunch — she didn't, so Dashon did not eat it.
Step 4: Clue 5: exactly one person ate it, so it was Wendy.

2 **Answer: B** $22 - 3x^2$

Step 1: Distribute: $3(4 - x^2) = 12 - 3x^2$.
Step 2: Add 10: $22 - 3x^2$.

3 **Answer: C** $1,024$

Step 1: From the table the count doubles every hour: $1, 2, 4, 8, \ldots$ so $y = 2^t$.
Step 2: After 10 hours: $2^{10} = 1024$.

4 **Answer: C** $66°$

Step 1: King St. $\parallel$ Monroe St., so the angles of the small triangle formed with Elk and Park Avenues are $71°$ and $43°$ (corresponding / alternate angles).
Step 2: The angles of a triangle sum to $180°$: $m\angle 3 = 180° - 71° - 43° = 66°$.

5 **Answer: B** $60°$

Step 1: Exterior angle theorem: $m\angle ABD = m\angle BDC + m\angle DCB$.
Step 2: Let $m\angle DCB = x$: $2x = 30 + x \Rightarrow x = 30°$.
Step 3: $m\angle ABD = 2(30°) = 60°$.

6 **Answer: C** $4\sqrt{5}$

Step 1: Radius $= 2 + 4 = 6$ cm; the chord is 4 cm from the center.
Step 2: A radius perpendicular to a chord bisects it: half-chord $= \sqrt{6^2 - 4^2} = \sqrt{20} = 2\sqrt{5}$.
Step 3: Chord $= 2 \times 2\sqrt{5} = 4\sqrt{5}$ cm.

7 **Answer: B** $5 \cdot 3^8$

Step 1: $\sqrt{25 \cdot 3^{16}} = \sqrt{25} \cdot \sqrt{3^{16}}$.
Step 2: $= 5 \cdot 3^8$.

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8 **Answer: A** $2x + y = 3$

Step 1: Point-slope: $y + 5 = -2(x - 4)$.
Step 2: $y + 5 = -2x + 8 \Rightarrow 2x + y = 3$.

9 **Answer: C** $\sqrt{24}$

Step 1: PQ is a diameter: $PQ = 2 \times \frac{5}{2} = 5$.
Step 2: An angle inscribed in a semicircle is a right angle, so $\angle PRQ = 90°$.
Step 3: $PR^2 = 5^2 - 1^2 = 24 \Rightarrow PR = \sqrt{24}$.

10 **Answer: B** Inverse property of multiplication

Step 1: In step 4, 3^2 is multiplied by $\frac{1}{3^2}$ to get 1.
Step 2: A number times its reciprocal equals 1: the inverse property of multiplication.

11 **Answer: A** $(x - 2)^2 + (y - 4)^2 = 13$

Step 1: Center = midpoint of the diameter: $\left(\frac{-1+5}{2}, \frac{6+2}{2}\right) = (2, 4)$.
Step 2: $r^2 = (5 - 2)^2 + (2 - 4)^2 = 9 + 4 = 13$.
Step 3: $(x - 2)^2 + (y - 4)^2 = 13$.

12 **Answer: C** 9

Step 1: Lengths scale by 3, so areas scale by 3^2.
Step 2: The larger poster's area is 9 times the smaller one's.

13 **Answer: A** Event 1: Toss a coin. Event 2: Toss the same coin again.

Step 1: Independent events: the first outcome does not change the probability of the second.
Step 2: Tossing a coin twice — the coin has no memory. (Drawing without replacement, rain/wet sidewalk, etc. are dependent.)

14 **Answer: D** $x = 5$ or $x = \frac{1}{3}$

Step 1: $2|3x - 8| = 14 \Rightarrow |3x - 8| = 7$.
Step 2: $3x - 8 = 7 \Rightarrow x = 5$; $3x - 8 = -7 \Rightarrow x = \frac{1}{3}$.
Step 3: Solutions: $x = 5$ and $x = \frac{1}{3}$.

15 **Answer: C** $\frac{2x\sqrt{6}}{9}$

Step 1: $\frac{\sqrt{8x^3}}{\sqrt{27x}} = \sqrt{\frac{8x^2}{27}} = \frac{2\sqrt{2}x}{3\sqrt{3}}$.
Step 2: Rationalize: $\frac{2\sqrt{2}x}{3\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{2\sqrt{6}}{9}x$.

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16 **Answer: A** $(2x-3y)^2$

Step 1: $4x^2=(2x)^2$, $9y^2=(3y)^2$, and $12xy=2(2x)(3y)$.
Step 2: Perfect square: $(2x-3y)^2$.

17 **Answer: C** $2+3\sqrt{2}$

Step 1: $\triangle EDC$: isosceles right with legs $DE=DC=1$, so $EC=\sqrt{2}$.
Step 2: Square $ABCE$ has sides $\sqrt{2}$.
Step 3: Perimeter of $ABCDE$ (not counting the inside segment EC): $AB+BC+CD+DE+EA=\sqrt{2}+\sqrt{2}+1+1+\sqrt{2}=2+3\sqrt{2}$.

18 **Answer: B** $a>0$

Step 1: $x^2=2a$.
Step 2: Two distinct real roots need $2a>0$, i.e. $a>0$ (then $x=\pm\sqrt{2a}$).

19 **Answer: A** $h=\frac{2A}{b}$

Step 1: $A=\frac{1}{2}bh \Rightarrow 2A=bh$.
Step 2: Divide by b: $h=\frac{2A}{b}$.

20 **Answer: A** 18 ft

Step 1: The fence and the house wall are parallel, so the small and large right triangles are similar.
Step 2: Distance from the ladder's foot: fence 1.5 ft, house $1.5+3=4.5$ ft (3 times as far).
Step 3: Height on the house $=3\times 6=18$ ft.

21 **Answer: A** If two lines cut by a transversal form a pair of congruent alternate interior angles, then the two lines are parallel.

Step 1: From the congruent triangles, $\angle A\cong\angle E$; these are alternate interior angles for $\overline{AB}$ and $\overline{EF}$ with transversal $\overline{AE}$.
Step 2: Lines cut by a transversal with congruent alternate interior angles are parallel.

22 **Answer: A** Maple and 6th

Step 1: Each block is $\frac{1}{4}$ mile, so the siren reaches corners up to 2 blocks away.
Step 2: Maple and 6th is the central corner — every corner of Newton is within 2 blocks of it.

23 **Answer: D** 27

Step 1: Similar solids with scale factor 3 have volume ratio $3^3=27$.
Step 2: The larger planter needs 27 bags.

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24 **Answer: C** $-4x^2 + 5x + 6$

Step 1: Distribute the minus sign: $x^2 + 4x + 4 - 5x^2 + x + 2$.
Step 2: Combine like terms: $-4x^2 + 5x + 6$.

25 **Answer: B** $64x^{10}$

Step 1: $(\sqrt{16x^6})^2 = 16x^6$ and $(\sqrt{4x^4})^2 = 4x^4$.
Step 2: Product: $64x^{10}$.

26 **Answer: C** $\frac{9\sqrt{2}}{4}$

Step 1: $162 = 16t^2 \Rightarrow t^2 = \frac{162}{16} = \frac{81}{8}$.
Step 2: $t = \frac{9}{\sqrt{8}} = \frac{9}{2\sqrt{2}} = \frac{9\sqrt{2}}{4}$ seconds.

27 **Answer: A** The diagonals of a rectangle are congruent.

Step 1: $\overline{MO}$ goes from the center to the circle, so $MO = 2$ cm (a radius).
Step 2: $\overline{PN}$ is the other diagonal of rectangle $MNOP$, and a rectangle's diagonals are congruent, so $PN = 2$ cm.

28 **Answer: D** 9

Step 1: $f(-2) = (-2)^2 - 3(-2) - 1$.
Step 2: $= 4 + 6 - 1 = 9$.

29 **Answer: A** Line with y-intercept 6 and slope 2

Step 1: "The first member (x) is 3 less than half the second (y)": $x = \frac{1}{2}y - 3$.
Step 2: Solve for y: $y = 2x + 6$ — slope 2, y-intercept 6.
Step 3: Graph A is the steep rising line crossing the y-axis at 6.

30 **Answer: C** 6 in, 7 in, 12 in

Step 1: With longest side c: obtuse when $a^2 + b^2 < c^2$ (and $a + b > c$).
Step 2: $6^2 + 7^2 = 85 < 12^2 = 144$, and $6 + 7 > 12$, so $6, 7, 12$ is obtuse. (5-12-13 is right; the others are acute.)

31 **Answer: C** $y = -\frac{5}{2}x + 3$

Step 1: Perpendicular slope: negative reciprocal of $\frac{2}{5}$ is $-\frac{5}{2}$.
Step 2: Through $(2, -2)$: $y + 2 = -\frac{5}{2}(x - 2) = -\frac{5}{2}x + 5$.
Step 3: $y = -\frac{5}{2}x + 3$.

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32 **Answer: B** $55°$

Step 1: A tangent is perpendicular to the radius: $m\angle HYX = 90°$.
Step 2: The exterior angle at X is $145°$, so $m\angle HXY = 35°$.
Step 3: $m\angle XHY = 180° - 90° - 35° = 55°$.

33 **Answer: C** $400\ \text{ft}^2/\text{gal}$

Step 1: Rate of change = slope = change in area ÷ change in gallons.
Step 2: From the graph, 1 gallon covers $400\ \text{ft}^2$: $\frac{400-0}{1-0} = 400\ \text{ft}^2/\text{gal}$.

34 **Answer: A** $1.42

Step 1: Let p = postcard price, e = envelope price: $14p + 5e = 12$ and $10p + 15e = 24.80$.
Step 2: Multiply the first by 3: $42p + 15e = 36$; subtract the second: $32p = 11.20 \Rightarrow p = 0.35$.
Step 3: $5e = 12 - 14(0.35) = 7.10 \Rightarrow e = 1.42$, i.e. $1.42.

35 **Answer: C** Blue

Step 1: When the net folds, the face two squares away in a straight line from red becomes its opposite face.
Step 2: Opposite faces never touch; that face is blue, so blue is not adjacent to red.

36 **Answer: A** $f(x) = -4x^2 - 56x - 202$

Step 1: Expand: $(x+7)^2 = x^2 + 14x + 49$.
Step 2: $-4(x^2 + 14x + 49) - 6 = -4x^2 - 56x - 196 - 6$.
Step 3: $= -4x^2 - 56x - 202$.

37 **Answer: B** $7x - 5$

Step 1: Find two numbers with product $21 \times (-20) = -420$ and sum 13: 28 and -15.
Step 2: $21x^2 + 28x - 15x - 20 = 7x(3x+4) - 5(3x+4) = (3x+4)(7x-5)$.
Step 3: So $7x - 5$ is a factor.

38 **Answer: D** View Set D

Step 1: Look at the solid from the front, from the right side, and from above, and record the visible squares in each direction.
Step 2: Only view set D matches all three outlines of the figure.

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39 **Answer: C** $a_n = 6n - 18$

Step 1: The terms increase by 6: arithmetic with $d = 6$.
Step 2: $a_n = a_3 + (n-3)d = 0 + 6(n-3) = 6n - 18$.

40 **Answer: A** $28°$

Step 1: In right $\triangle ABC$: $\cos C = \frac{BC}{AC} = \frac{15}{17}$, so $m\angle C \approx 28.1°$.
Step 2: Similar triangles: $m\angle F = m\angle C \approx 28°$.

41 **Answer: B** $x + 3y = 9$

Step 1: Slope $= \frac{6-2}{-9-3} = -\frac{1}{3}$.
Step 2: $y - 2 = -\frac{1}{3}(x - 3) \Rightarrow 3y - 6 = -x + 3 \Rightarrow x + 3y = 9$.

42 **Answer: B** $g(x) = \frac{1}{3}f(x)$

Step 1: Compare y-values at the same x: every y-value of g is one third of f's (same x-intercept, slope divided by 3).
Step 2: A vertical compression by $\frac{1}{3}$: $g(x) = \frac{1}{3}f(x)$.

43 **Answer: B** $4\sqrt{3}$

Step 1: $DC = 12 - 8 = 4$.
Step 2: Leg rule for the altitude to the hypotenuse: $BC^2 = DC \cdot AC = 4 \times 12 = 48$.
Step 3: $BC = 4\sqrt{3}$.

44 **Answer: C** $(11, 2\sqrt{12})$

Step 1: Radius$^2 = (-5)^2 + 12^2 = 169$.
Step 2: $(11)^2 + (2\sqrt{12})^2 = 121 + 48 = 169$ ✓ — the other points give 109, 313, 589.

45 **Answer: A** If tomorrow is not Saturday, then today is not Friday.

Step 1: Contrapositive of "if p, then q" is "if not q, then not p".
Step 2: p: today is Friday; q: tomorrow is Saturday.
Step 3: "If tomorrow is not Saturday, then today is not Friday."

46 **Answer: B** The coordinates of the y-intercept are $(0, 8)$

Step 1: At $x = 0$: $y = 8(0.25)^0 = 8$, so the y-intercept is $(0, 8)$.
Step 2: (An exponential $8(0.25)^x$ never reaches 0 — no x-intercept; its asymptote is $y = 0$; at $x = 2$, $y = 0.5$.)

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47 **Answer: D** Graph D

Step 1: A graph is a function if every vertical line meets it at most once.
Step 2: Only graph D passes the vertical line test.

48 **Answer: A** 1 Point, 2 line

Step 1: Place the two congruent semicircles on the same line, facing each other along the line.
Step 2: The figure has 2 lines of symmetry (the common line and the perpendicular bisector between them) and 1 point of symmetry (the center).

49 **Answer: D** Line with y-intercept 3 and x-intercept -3

Step 1: "The first member (x) is 3 less than the second (y)": $x = y - 3 \Rightarrow y = x + 3$.
Step 2: y-intercept: $(0, 3)$; x-intercept: $(-3, 0)$ — graph D.

50 **Answer: B** $\angle G \cong \angle T$

Step 1: Correspondence from the names: $J \leftrightarrow M, G \leftrightarrow S, R \leftrightarrow T$.
Step 2: $\angle G \cong \angle T$ pairs non-corresponding angles, so it is not always true.

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